E-Commerce Customer Segmentation

RFM Analysis with K-Means Clustering

Project Report

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Item	Detail
Technique	RFM (Recency, Frequency, Monetary) + K-Means
Tools	Python (Pandas, Scikit-learn, Matplotlib, Seaborn)
Dataset	Online-retail style transactions, Dec 2010 - Dec 2011
Customers analysed	600
Transactions (clean)	13,394

Executive Summary

Customers were segmented by how recently they purchased, how often, and how much they spent. Four business segments emerged. **Champions** are only 22% of customers but generate **68.9%** of revenue. The **At-Risk** segment (241 customers, 26.9% of revenue) used to buy about 6 times each but has been silent for ~117 days; it is the best target for a win-back campaign.

1. Data and Cleaning

The raw file has **14,383** rows and 8 columns (InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, Country). The following cleaning steps were applied:

- Removed rows with missing CustomerID (814 rows, guest checkouts) since RFM needs a customer identity.
- Removed cancelled orders (InvoiceNo starting with 'C', 176 rows).
- Removed rows with Quantity or UnitPrice less than or equal to zero, and exact duplicates.
- Created TotalPrice = Quantity x UnitPrice.

After cleaning: **13,394** rows across **600** customers.

2. Methodology

2.1 Building the RFM table

A snapshot date (last transaction date + 1 day) was used as 'today'. For each customer:

- **Recency**: days since the last purchase (lower is better).
- **Frequency**: number of unique invoices (higher is better).
- **Monetary**: total spend (higher is better).

2.2 RFM scores

Each metric was split into quintiles and scored 1-5, where 5 is best (Recency labels are reversed). A score of 555 is the best customer and 111 the weakest. A rule-based segmentation on R and F scores was also created as a business-friendly baseline.

2.3 K-Means clustering

Frequency and Monetary are heavily right-skewed, so a log transform (log1p) was applied, followed by standardisation so that no single metric dominates the distance calculation. The number of clusters was chosen using the Elbow method and Silhouette score; K = 4 was selected (silhouette = 0.45). Clusters were then named by reading each cluster's average R, F and M values, so that the names make sense to a marketing manager.

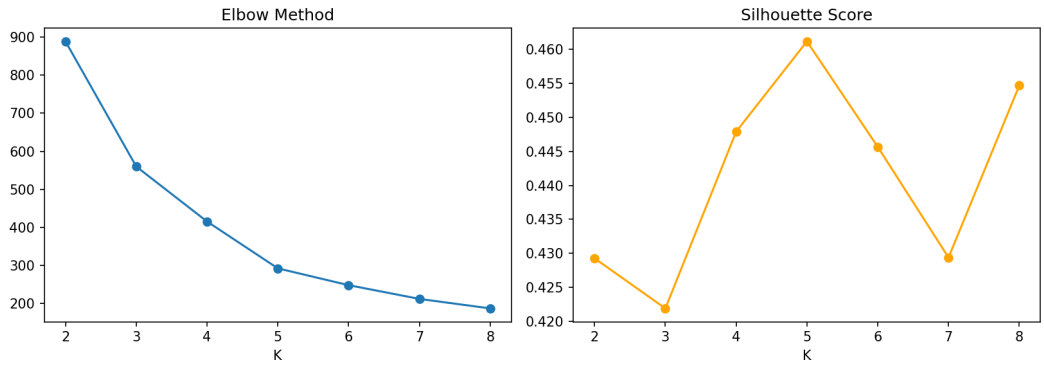


Figure 1: Elbow and Silhouette analysis for choosing K

3. Results: Customer Segments

Segment	Customers	Avg Recency (days)	Avg Orders	Avg Spend	Total Revenue	Revenue %
Champions	131	7.5	15.1	2,539	332,625	68.9%
At-Risk	241	116.9	6.4	538	129,722	26.9%
New Customers	92	32.7	1.4	86	7,862	1.6%
Lost	136	288.6	1.5	92	12,446	2.6%

Segment definitions

- **Champions**: bought very recently, order often and spend the most.
- **At-Risk**: previously frequent, decent spenders, but have not purchased for months.
- **New Customers**: purchased recently but only once or twice so far; low spend yet.
- **Lost**: long inactive, few orders and low spend.

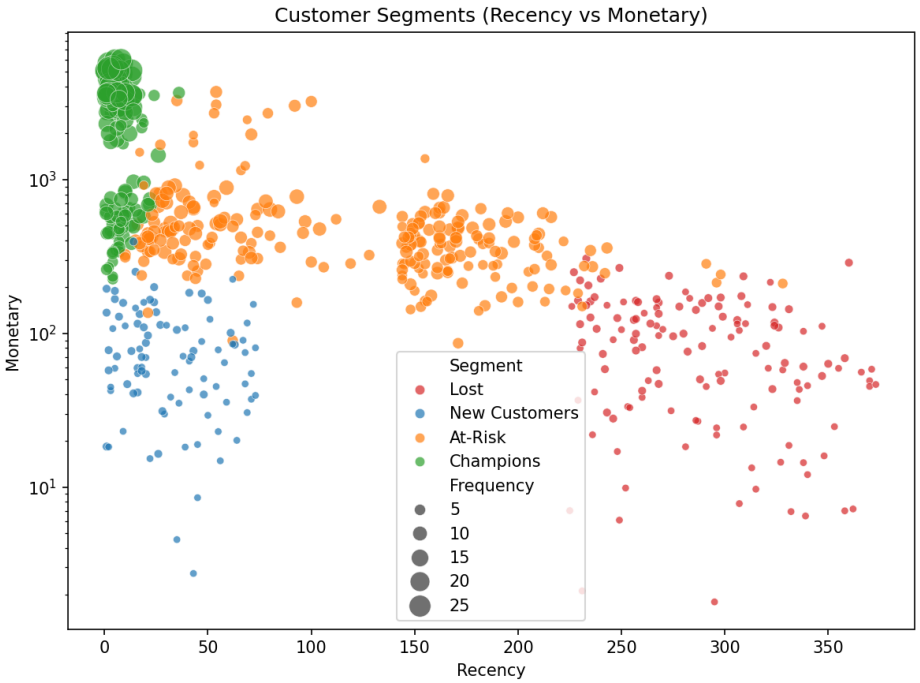


Figure 2: Segments by Recency vs Monetary (bubble size = Frequency, log scale on Monetary)

4. Segment Size vs Revenue

The gap between customer count and revenue share is the key business insight: a small group of customers carries most of the revenue.

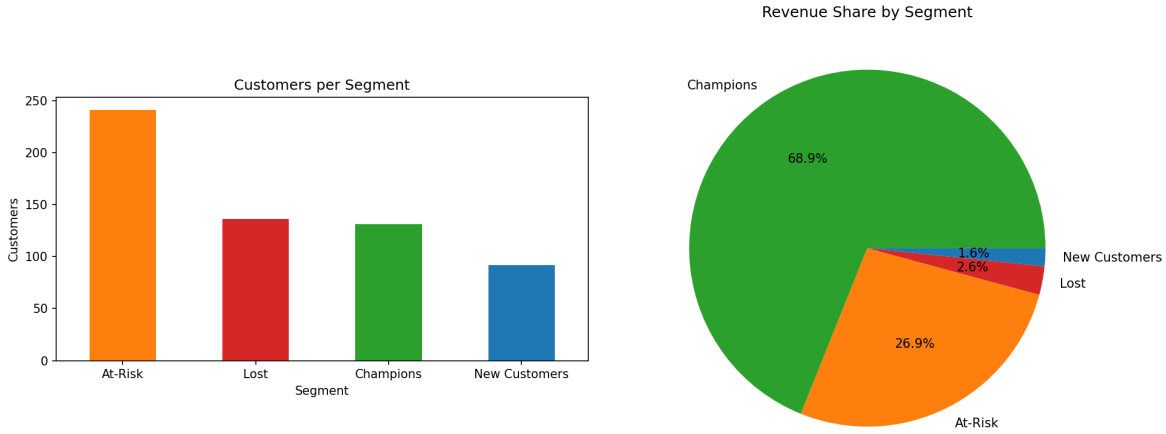


Figure 3: Customers per segment (left) and revenue share by segment (right)

Key observations

- Champions: 131 customers (22%) produce 68.9% of revenue.
- At-Risk: 241 customers (40%) still account for 26.9% of revenue, so losing them would hurt.
- Lost: 136 customers (23%) contribute only 2.6% of revenue.
- New Customers: 92 customers contribute 1.6%, but they are a pipeline for future growth.

5. Business Recommendation

Primary target: At-Risk segment (241 customers, 26.9% of revenue)

Campaign: Win-back - a personalised 'We miss you' email/SMS with a 10-15% discount or free shipping, valid for 7 days, featuring products they bought before.

Why this segment?

- They have proven loyalty (about 6 orders on average) and meaningful spend (~538 each).
- They have been inactive for ~117 days: slipping, but not yet gone. Win-back chances are far higher than for Lost customers.
- A modest discount costs far less than acquiring a brand-new customer.

Action plan for every segment

Segment	Priority	Recommended action
At-Risk	High	Win-back offer (10-15% off, 7-day validity), personalised product picks
Champions	High (retain)	Loyalty rewards, early access to new products, referral programme; avoid heavy discounts
New Customers	Medium	Welcome series, second-purchase incentive to build the habit
Lost	Low	One cheap automated reactivation email; otherwise do not spend budget

Success metrics

- Reactivation rate of At-Risk customers (target: purchase within 30 days of the campaign).
- Incremental revenue vs a holdout group that receives no offer (A/B test).
- Repeat-purchase rate of New Customers after the welcome series.

6. Conclusion and Limitations

RFM combined with K-Means turns raw transactions into four clear, actionable customer groups. Revenue is highly concentrated, so retaining Champions and winning back At-Risk customers is the best use of marketing budget. **Note:** this analysis used a synthetic, retail-style dataset built for practice; absolute numbers will differ on real data (e.g. the UCI Online Retail dataset), but the method and code are identical. RFM also ignores product mix and margin, which could be added in future work.